

Name.....

Roll No.....

ST. JOSEPH'S COLLEGE OF ENGINEERING AND TECHNOLOGY, PALAI.

(An ISO 9001:2015 Certified College)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**B. TECH DEGREE FIRST INTERNAL TEST- MARCH 2024****Eighth Semester****CST402 DISTRIBUTED COMPUTING**

Time: 1.5 Hours

Max. Marks: 50

PART A

Answer all questions (3 x 6 = 18 Marks)

Qn. No.	Question	Marks	K level of Qn.	CO & KL
1	List the features of the Distributed System	3	K2	CO1 & K2
2	What are the basic properties of scalar time.	3	K2	CO1 & K2
3	Explain about Termination Detection.	3	K2	CO2 & K3
4	What is the significance of global state in a distributed system environment?	3	K2	CO2 & K3
5	Differentiate between Logical and Physical concurrency	3	K2	CO1 & K2
6	Explain the concept of "cut" in the context of distributed computing.	3	K2	CO1 & K2

PART B

Answer any two questions (8 x 2 = 16 Marks)

Qn. No.	Question	Marks	K level of Qn.	CO & KL
7	Explain the design issues of a distributed system.	8	K2	CO1 & K2
8	Discuss about various primitives for distributed communication	8	K2	CO1 & K2
9	What are the applications of Distributed Computing.	8	K2	CO1 & K2

PART C

Answer any two questions (8 x 2 = 16 Marks)

Qn. No.	Question	Marks	K level of Qn.	CO & KL
10	Illustrate the Working of Spanning Tree based Termination Detection Algorithm.	8	K3	CO2 & K3
11	Illustrate bully algorithm for electing a new leader.	8	K2	CO2 & K3
12	Explain in detail about Chandy Lamport algorithm	8	K2	CO2 & K3

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Time: 1.5 Hours

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PART A

Answer all questions (3 x 6 = 18 Marks)

Qn. No.	Question	Marks	K level of Qn.	CO & KL
1	List the requirements of Mutual Exclusion Algorithm	3	K1	CO3 & K2
2 ✓	How do we utilize the Wait-for-Graph (WFG) to identify deadlock within a distributed environment?	3	K2	CO4 & K2
3 ✓	Describe the challenges associated with Deadlock Detection in the Distributed Systems	3	K1	CO4 & K2
4 ✓	What metrics can be used to measure the performance of mutual exclusion algorithms?	3	K2	CO4 & K2
5 ✓	Define Byzantine agreement problem	3	K1	CO5 & K2
6	Write the features of SUN Network File System	3	K1	CO6 & K2

PART B

Answer any two questions (8 x 2 = 16 Marks)

Qn. No.	Question	Marks	K level of Qn.	CO & KL
7 ✓	Illustrate the Ricart-Agrawala Algorithm for Mutual Exclusion using an example.	8	K2	CO3 & K2
8 ✓	Compare various models of deadlock in the distributed systems	8	K2	CO4 & K2
9	Explain in Detail about Deadlock handling Strategies in a Distributed environment	8	K2	CO4 & K2

PART C

Answer any two questions (8 x 2 = 16 Marks)

Qn. No.	Question	Marks	K level of Qn.	CO & KL
10 ✓	Explain consensus algorithm for crash failures under synchronous systems	8	K1	CO5 & K3
11 ✓	Explain google file system architecture.	8	K1	CO6 & K3
12	Explain SUN -NFS Architecture.	8	K1	CO6 & K3